

Comparison of Artificial Intelligence-Based Machine Learning Classifiers for Early Detection of Keratoconus

European Journal of Ophthalmology
2022, Vol. 32(3) 1352–1360
© The Author(s) 2022
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/11206721211073442
journals.sagepub.com/home/ejo



Mehrdad Mohammadpour^{1,3} , Zahra Heidari^{2,3} ,
Hassan Hashemi³ , Mehdi Yaseri⁴ and Akbar Fotouhi⁴

Abstract

Purpose: To compare the agreement between artificial intelligence (AI)-based classifiers and clinical experts in categorizing normal cornea from ectatic conditions.

Methods: Prospective diagnostic test study at Noor Eye Hospital. Two hundred twelve eyes of 212 patients were categorized into three groups of 92 normal, 52 subclinical keratoconus (SKCN), and 68 KCN eyes based on clinical findings by 3 independent expert examiners. All cases were then categorized using four different classifiers: Pentacam Belin/Ambrosio enhanced ectasia total deviation value (BADD) and Topographic Keratoconus Classification (TKC), Sirius Phoenix, and OPD-Scan III Corneal Navigator. The performance of classifiers and their agreement with expert opinion were investigated using the sensitivity, specificity, and Kappa index (κ).

Results: For detecting SKCN, Phoenix had the highest agreement with the clinical diagnosis (sensitivity, specificity, and κ of 84.62%, 90.0%, and 0.70, respectively) followed by BADD (55.56%, 86.08%, 0.42), TKC (26.92%, 97.50%, 0.30), and Corneal Navigator (30.77%, 93.75%, 0.29). For KCN diagnosis, the highest agreement with expert opinion was seen for Phoenix (80.02%, 96.60%, 0.79), BADD (95.59%, 85.42%, 0.75), TKC (95.59%, 84.03%, 0.73), and Corneal Navigator (67.65%, 96.45%, 0.68). Analysis of different classifiers showed that Phoenix had the highest accuracy for differentiating KCN (91.24%) and SKCN (88.68%) compared to other classifiers.

Conclusions: Although AI-based classifiers, especially Sirius Phoenix, can be very helpful in detecting early keratoconus, they cannot replace clinical experts' opinions, particularly for decision-making before refractive surgery. Albeit, there may be concerns about the accuracy of clinical experts as well.

Keywords

Keratoconus, subclinical keratoconus, artificial intelligence, corneal topography

Date received: 18 July 2021; accepted: 12 December 2021

Introduction

Keratoconus (KCN) is a progressive corneal ectasia that can be easily diagnosed during routine clinical examinations in its advanced stages, but diagnosing subclinical KCN (SKCN) is a diagnostic challenge when clinical signs are still absent. Early diagnosis of keratoconus is important for a timely intervention to stop or slow the progression of the disease and maintain good visual acuity.¹

In recent years, advances in diagnostic and screening modalities and access to ophthalmological care through teleophthalmology has made artificial intelligence (AI) an important assistive tool for the timely diagnosis and

¹Department of Ophthalmology, Farabi Eye Hospital and Eye Research Center, Faculty of Medicine, Tehran University of Medical Sciences, Tehran, Iran

²Department of Rehabilitation Science, School of Allied Medical Sciences, Mazandaran University of Medical Sciences, Sari, Iran

³Noor Ophthalmology Research Center, Noor Eye Hospital, Tehran, Iran

⁴Department of Epidemiology and Biostatistics, School of Public Health, Tehran University of Medical Sciences, Tehran, Iran

Corresponding author:

Zahra Heidari, Department of Rehabilitation Science, School of Allied Medical Sciences, Mazandaran University of Medical Sciences, Sari, Iran.
Email: zahra.heidari77@yahoo.com

management of preventable or treatable eye diseases and appropriate follow-up of patients.^{2,3} AI technology converts unstructured data into structured data to distinguish normal from pathological conditions. Machine learning (ML) algorithms are one of the AI techniques that complement diagnostic and therapeutic strategies using complex data derived from imaging clinical features and classifying the parameters based on the training database.⁴ Various algorithms used in ML include decision tree,⁵ neural networks,^{6,7} linear discriminant analysis,⁸ and support vector machine.⁹

Different corneal imaging systems with advanced AI-based classifiers are available for the diagnosis and classification of at-risk corneas and KCN eyes. In Placido disk-based topography, the curvature of the cornea is calculated after measuring the slope on images taken from the anterior corneal surface and the reflected rings. OPD Scan III (Nidek Co. Ltd, Gamagori, Japan) is a Placido disc topographer that presents a topographic classifier map using the Corneal Navigator function and neural network processing.⁶

Scheimpflug-based systems can evaluate a more extensive surface area of the cornea compared to Placido systems through three-dimensional imaging.¹⁰ Pentacam (Oculus Optikgeräte GmbH, Wetzlar, Germany) uses a rotating Scheimpflug camera to take at least 25 images during 2 s and calculates tomographic information from the anterior and posterior corneal surfaces and full pachymetric data.¹¹ Topographic Keratoconus Classification (TKC) and the Belin/Ambrosio enhanced ectasia total deviation value (BADD) are two Pentacam classifiers which are derived from regression algorithms.^{12,13}

Some modern systems combine Placido and Scheimpflug technologies to provide complete information on the anterior and posterior corneal surfaces. Sirius (Costruzione Strumenti Oftalmici, Florence, Italy) takes 25 Scheimpflug images and 1 Placido image in less than one second. Height, slope, and curvature data are then calculated with an arc-step method. This system provides comprehensive information on the entire cornea and classifies keratoconus via the Phoenix software through a neural network process.¹⁴

The various AI-based KCN classifiers can be helpful to prevent postoperative ectasia in corneal refractive surgery. To the best of our knowledge, there are few studies on the agreement between these classifiers and expert opinion in differentiating normal cornea from keratoconic conditions. Therefore, this study was conducted to investigate the diagnostic accuracy and agreement of four device-based classifiers (BADD and TKC, Phoenix software, and Corneal Navigator) with expert opinion in detecting normal, SKCN, and KCN eyes.

Materials and methods

The protocol of this prospective study was approved by the Ethics Committee of Tehran University of Medical

Sciences (IR.TUMS.VCR.REC.1396.4815), and the study was conducted at Noor Eye Hospital, Tehran, Iran. All participants gave written informed consent, and all study procedures were in accordance with the Declaration of Helsinki.

Study groups

Two hundred and seventeen eyes of 212 patients aged 17–49 years who were referred to the Keratoconus Clinic or were refractive surgery candidates at the Refractive Surgery Unit were included in this prospective diagnostic test study. The exclusion criteria were a history of ocular surgery, corneal cross-linking, or ring implantation; corneal hydrops or scarring; signs and symptoms of dry eye or ocular diseases other than keratoconus; connective tissue diseases; systemic diseases affecting the eyes; corneal haze; pregnancy; and contact lens use in the previous one month.

All patients underwent corneal imaging using the Pentacam (Oculus, Wetzlar, Germany), Sirius (Costruzione Strumenti Oftalmici, Italy), and OPD-Scan III (Nidek Co. Ltd, Gamagori, Japan), and only one eye was randomly selected and analyzed. Measurements were repeated for each patient until images with an “OK” quality statement (good quality score) were recorded. Then patients were assigned to one of the following three groups based on the clinical diagnosis of two independent experienced corneal specialists (M. Mohammadpour, K. Amanzadeh), through slit-lamp biomicroscopy, retinoscopy, corrected distance visual acuity (CDVA) measurement with a Snellen chart, and evaluation of the Pentacam Refractive 4 Maps (Figure 1); they were blinded to classification reports. Cases with diagnostic discrepancy were resolved by a third expert examiner (A. Moghaddasi) for a definitive diagnosis.

1) Normal: Refractive surgery candidates with completely normal eyes in terms of slit-lamp biomicroscopy, retinoscopy, topographic, and tomographic criteria were assigned to the normal (N) comparison group.

2) SKCN: Eyes without scissoring on retinoscopy and no biomicroscopic sign of KCN (such as Vogt striae, Fleischer ring, Munson’s sign, Rizzuti’s phenomenon), and CDVA equal to 20/20 Snellen (0.0 logMAR), and presence of abnormal topographic pattern including skewed asymmetric bow tie, central or inferior steepening, $10 < \text{skewed radial axes (SRAX)} < 20$ degrees, $47.2 < \text{Maximum keratometry (Kmax)} < 48.7$ diopters (D), and $1.4 < \text{inferior-superior (IS) value} < 1.9$ D at 6 mm (3 mm radii) were assigned to the SKCN group.^{15–17}

3) KCN: Eyes with scissoring reflex on retinoscopy and presence of at least one of the specific signs of KCN on clinical examination including Fleischer ring, Vogt striae, apical thinning, Munson’s sign or Rizzuti’s phenomenon, and CDVA less than 20/20 Snellen, and abnormal topographic

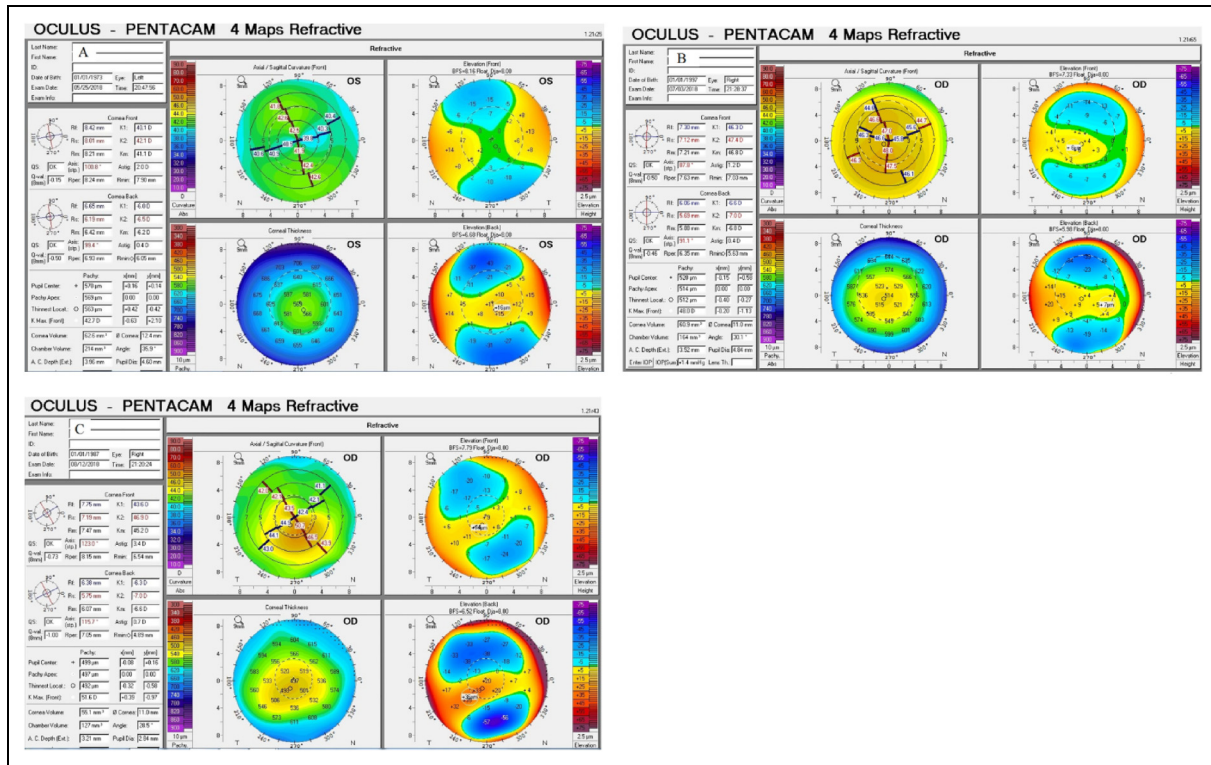


Figure 1. Refractive maps of normal (A), subclinical keratoconus (B), and clinical keratoconus (C) corneas.

findings including skewed asymmetric bow tie, central or inferior steepening or a claw pattern on topography, SRAX >21 degrees, Kmax > 48.7 D, and IS value >1.9 D at 6 mm (3 mm radii) were assigned to the KCN group.^{15,16,18}

Topographic and tomographic measurements

Pentacam (software version 1.22r03) provides TKC and BADD classifiers based on regression analysis. TKC is adapted from the classical Amsler-Muckenhirn standards¹³ and classifies eyes into grade 0 (normal) up to grade 4 (sever KCN), possible (SKCN), corneal post-surgery, and abnormal.

BADD is a screening tool that is calculated based on the analysis of pachymetric progression and posterior and anterior elevation maps including five indices of deviation of the difference map on the front/back surface (Df and Db), pachymetric progression (Dp), thinnest point (Dt), and deviation of the Ambrosio Relational Thickness parameter (Da).¹² Each D parameter value is reported as standard deviations (SD) from the mean. Optimal BADD (total final D) cut points, according to the Pentacam manual, are $D < 1.6 \text{ SD}$ for normal, $1.6 \text{ SD} \leq D < 3.0 \text{ SD}$ for SKCN, and $D \geq 3.0 \text{ SD}$ for KCN.

Sirius classification system (Phoenix software version 3.4.0.73) is a screening tool that classifies the cornea into normal, SKCN, KCN, and abnormal groups. This classification is defined based on corneal thickness, corneal

curvature and elevation, and aberrometry indices provided by the Phoenix software.

OPD-Scan III (version 1.03.02) Corneal Navigator uses 19 corneal indices, measurement similarity (0–99%) between the parameters, and the corneal shape to classify eyes into nine categories, including normal (NRM), astigmatism (AST), SKCN, KCN, Pellucid marginal degeneration (PMD), PKP (Penetrating Keratoplasty), MRS (Myopic Refractive Surgery), HRS (Hyperopic Refractive Surgery), or other (OTH).⁶ The OTH group includes corneas with unclassified variations. For better comparison with the classification parameters of other devices, regular astigmatism was considered normal. In this study, there was no case of PMD, PKP, or MRS; therefore, these groups are not shown in the results.

Figure 2 illustrates samples of the classification systems from Pentacam (A), Sirius (B), and (C) OPD-Scan III including TKC, BADD, Phoenix software analysis, and Corneal Navigator.

Statistics

Statistical analyses were performed using SPSS software (version 23; IBM Inc., New York, NY, USA). Descriptive statistics were summarized as the mean $\pm$ standard deviations (SDs) and ranges. To assess the accuracy of the AI-based classifiers compared to the expert opinion, the following parameters were calculated: true positive (TP), true negative (TN), false positive (FP),

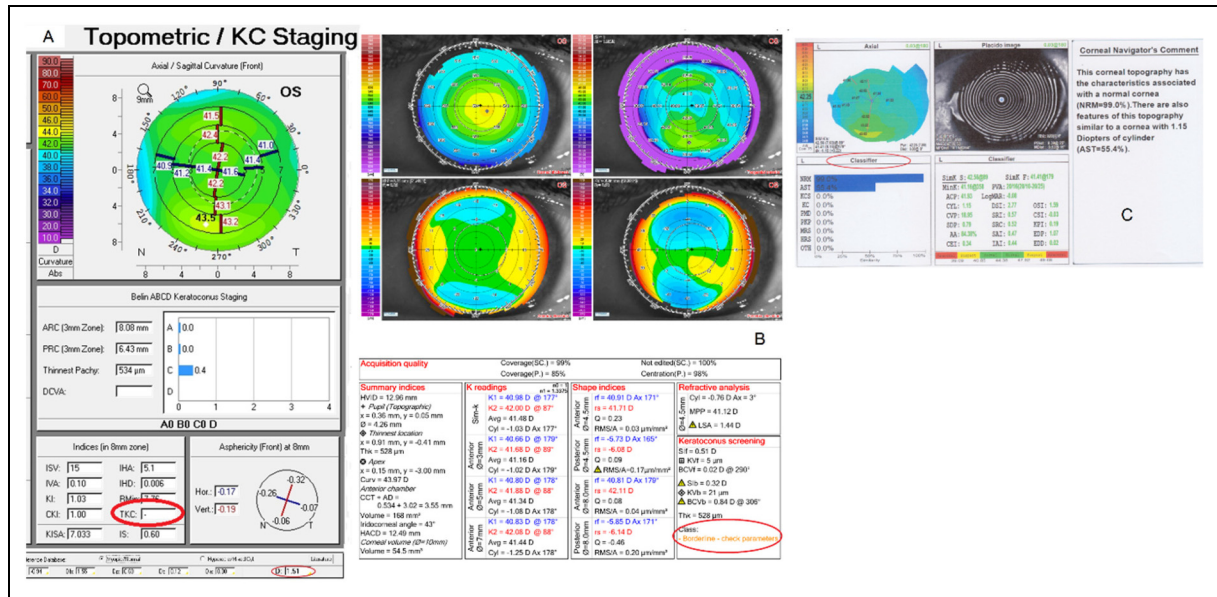


Figure 2. Samples of the classification systems from Pentacam (A), Sirius (B), and (C) OPD-Scan III including the BADD (Belin/Ambrosio enhanced ectasia total deviation value), TKC (Topographical Keratoconus Classification), Phoenix software analysis, and Corneal Navigator in a subclinical keratoconus eye.

Table 1. Cross Tabulations of Expert opinion and 4 classifiers by Pentacam, Sirius and OPD-Scan III in normal (N), subclinical keratoconus (SKCN) and keratoconus (KCN) groups.

Expert Opinion (Total = 212)	N (n = 92)	SKCN (n = 52)	KCN (n = 68)
Pentacam / BADD			
N (72)	68	4	0
SKCN (54)	21	30	3
KCN (86)	3	18	65
Pentacam / TKC			
N (103)	86	16	1
SKCN (19)	3	14	2
KCN (88)	1	22	65
Abnormal (2)	2	0	0
Sirius / Phoenix			
N (91)	85	5	1
SKCN (60)	4	44	12
KCN (59)	2	3	54
Abnormal (2)	1	0	1
OPD-Scan III / Corneal Navigator			
N (78)	63	13	2
KSC (26)	7	16	3
KC (51)	2	3	46
HRS (3)	2	1	0
OTH (52)	18	19	15

BADD = Belin/Ambrosio enhanced ectasia total deviation value, TKC = Topographical Keratoconus Classification of Pentacam, HRS = Hyperopic Refractive Surgery, OTH = Others (unclassified variations).

false negative (FN), sensitivity: $TP / (TP + FN)$; specificity: $TN / (TN + FP)$; positive likelihood ratio (LR^+): sensitivity / $1 - \text{specificity}$, negative likelihood ratio (LR^-): $1 - \text{sensitivity} /$

specificity, and diagnostic accuracy: $TP + TN / TP + TN + FP + FN$. The Cohen kappa index (κ) was calculated to evaluate the agreement between various classification

methods with expert opinion using the following formula: $P(o) - P(e) / 1 - P(e)$, where $P(o)$ is the proportion of observed agreement and $P(e)$ is the proportion expected by chance. This index ranges from -1.00 to $+1.00$, and a κ of 0.01 – 0.20 , 0.21 – 0.40 , and 0.41 – 0.60 , and 0.61 – 0.80 indicates poor, fair, moderate, and substantial agreement, respectively.¹⁹ For all estimated measures, 95% confidence intervals (CI) were computed as well.

Results

Two hundred-twelve eyes of 212 patients were included in the study. The age range of this sample was 17 to 49 years. There were 92 eyes in the normal group (mean age of 30.22 ± 6.74 years), 52 in the SKCN group (mean age of 30.50 ± 5.27 years), and 68 KCN eyes (mean age of 32.70 ± 6.05 years). There was no significant age difference between the normal and SKCN groups ($P = 1.00$); the age difference between the normal and KCN groups was significant ($P = 0.039$). Females comprised 63.04% ($n = 58$) of the normal group, whereas males accounted for 64.16% ($n = 77$) of the two keratoconic groups (SKCN + KCN).

The mean spherical equivalent (SE) was -2.91 ± 2.09 diopter (D) (range, -8.00 to $+7.13$), -2.39 ± 2.21 D (range, -9.25 to $+2.25$), and -4.14 ± 3.40 D (range, -18.38 to $+1.25$) in the normal, SKCN and KCN groups, respectively. The mean CDVA (logMAR) was 0.01 ± 0.05 (range, 0.0 to $+0.3$), 0.00 ± 0.00 (all logMAR 0.00), and 0.32 ± 0.21 (range, 0.1 to $+0.8$) in the normal, SKCN, and KCN groups, respectively. KCN cases had higher cylinder and lower CDVA compared to the normal group ($P < 0.001$). No significant differences were observed between the SKCN and the normal group in terms of refraction or visual acuity ($P = 0.98$).

With Pentacam, TKC classified a large number of normal (103/92, specificity 85.50%) and KCN eyes (88/68, sensitivity 95.59%, specificity 83.33%) correctly. However, this method had a weak sensitivity (26.92%) in differentiating SKCN from other groups as it classified 22/52 SKCN as KCN and 16/52 SKCN as N.

BADD diagnosed 68/92 N correctly (specificity 96.60%), but 21/92 N cases were classified as SKCN. BADD classified more than half (30/52) of the SKCN cases correctly (sensitivity 55.56%, specificity 86.08%); of the remaining SKCN cases 4 were classified as N and

Table 2. Comparison of the keratoconus classification with 4 classifiers and expert opinion in keratoconus (KCN) and subclinical keratoconus (SKCN) groups

	Pentacam		Sirius	OPD-Scan III
	BADD (95% CI)	TKC (95% CI)	Phoenix (95% CI)	Corneal Navigator (95% CI)
KCN (n = 68)				
TP	65	65	54	46
TN	123	120	139	136
FP	21	24	5	5
FN	3	3	14	22
Sensitivity	95.59% (87.81–98.49)	95.59 (87.81- 98.49)	80.0(69.18- 87.70)	67.65 (55.84- 77.56)
Specificity	85.42 (78.73- 90.26)	83.33 (76.40- 88.54)	96.6(92.29- 98.54)	96.45 (91.97- 98.48)
LR +	6.55 (5.92- 7.20)	5.73 (5.27–6.23)	23.52 (15.75–35.11)	19.08 (12.63–28.81)
LR-	0.05 (0.04- 0.10)	0.05 (0.02–0.10)	0.20 (0.17–0.23)	0.34 (0.31–0.37)
Diagnostic Accuracy	88.68 (83.71- 92.7)	87.26 (82.10- 91.10)	91.24 (86.73- 94.32)	87.08 (81.86- 90.97)
Kappa	0.75 (0.62- 0.88)	0.72 (0.59–0.86)	0.79 (0.66–0.92)	0.68 (0.55–0.80)
SKCN (n = 52)				
TP	30	14	44	16
TN	136	155	144	150
FP	24	5	16	10
FN	22	38	8	36
Sensitivity	55.56 (42.38- 68.0)	26.92 (16.77- 40.25)	84.62 (72.48- 91.99)	30.77 (19.91- 44.27)
Specificity	86.08 (77.82- 90.62)	97.50 (93.7- 99.02)	90.0 (84.37- 93.75)	93.75 (88.88- 96.57)
LR ⁺	3.99 (3.46- 4.59)	8.61 (3.98–18.64)	8.46 (7.42–9.64)	4.9 (3.07–7.89)
LR [–]	0.51 (0.47–0.56)	0.75 (0.71– 0.79)	0.17 (0.13–0.21)	0.73 (0.69–0.78)
Diagnostic Accuracy	78.30 (72.28- 83.32)	79.70 (73.80- 84.85)	88.68 (83.71- 92.27)	78.3 (72.28- 83.32)
Kappa	0.42 (0.28- 0.55)	0.30 (0.18–0.41)	0.70 (0.57–0.84)	0.29 (0.17–0.41)

TP = true positive, TN = true negative, FP = false positive, FN = false negative, LR^+ = likelihood ratio of a positive test, LR^- = likelihood ratio of a negative test, BADD = Total deviation of Belin/Ambrósio Enhanced Ectasia Display, TKC = Topographical keratoconus classification of Pentacam, CI = Confidence Interval.

18 as KCN. For KCN cases, BADD classified 65/68 correctly (sensitivity 95.59%, specificity 85.42%).

Sirius Phoenix differentiated normal eyes (85/92) from keratoconic eyes (specificity 95.0%). Most SKCN (44/52) (sensitivity 84.62%, specificity 90.0%) and KCN cases (54/68) (sensitivity 80.0%, specificity 96.6%) were classified correctly, although it classified 5/52 SKCN as N and 12/68 KCN as SKCN.

OPD-Scan III Corneal Navigator correctly classified 63/92 of the normal (specificity 87.50%) and 46/68 of the KCN cases (sensitivity 67.65%, specificity 96.45%), but for SKCN, it differentiated 16/52 cases correctly (sensitivity 30.77%, specificity 93.75%), and 13/52 SKCN cases were classified as N. Corneal Navigator classified some cases as OTH (18 N, 19 SKCN, and 15 KCN).

Cross-tabulations of expert opinion and the 4 studied classifiers are shown in Table 1.

In terms of accuracy for differentiating KCN from other groups, analysis of different classifiers showed that Phoenix had the highest accuracy (91.24%), followed by BADD (88.68%), TKC (87.71%), and Corneal Navigator (87.08%). For the diagnosis of SKCN, Phoenix also had the highest accuracy (88.68%) (Table 2).

All classifiers showed substantial agreement with clinical expert opinion in differentiating KCN from other groups, among which Phoenix and BADD had the highest agreement ($\kappa = 0.79$ and 0.75 , respectively).

In differentiating SKCN from other groups, except Phoenix and BADD ($\kappa = 0.70$ and 0.42 , respectively), other classifiers had a relatively fair agreement with expert clinical judgment ($\kappa < 0.40$) (Table 2).

Figure 3 shows the distribution of cases in each group by each classification method and clinical expert assessment.

Discussion

Given the importance of screening patients before refractive surgery, diagnostic methods based on machine learning algorithms and different classification systems can help identify the early forms of keratoconus. Multiple studies in the literature have looked at ways to distinguish between normal and keratoconic corneas using single or integrated parameters, however, due to the similarities between the clinical results between normal and SKC eyes, it is difficult to distinguish between the two groups. This causes the overlap of descriptive indices and the problems in differential analysis. Thus, it is important to use a combination of indices using AI-based analysis. These algorithms objectively classify the eyes by combining parameters in different corneal imaging systems.

For Pentacam, TKC and BADD classification systems were evaluated. The result of this study showed that both classifiers had substantial agreement with the clinical expert for detecting definite KCN. BADD was able to

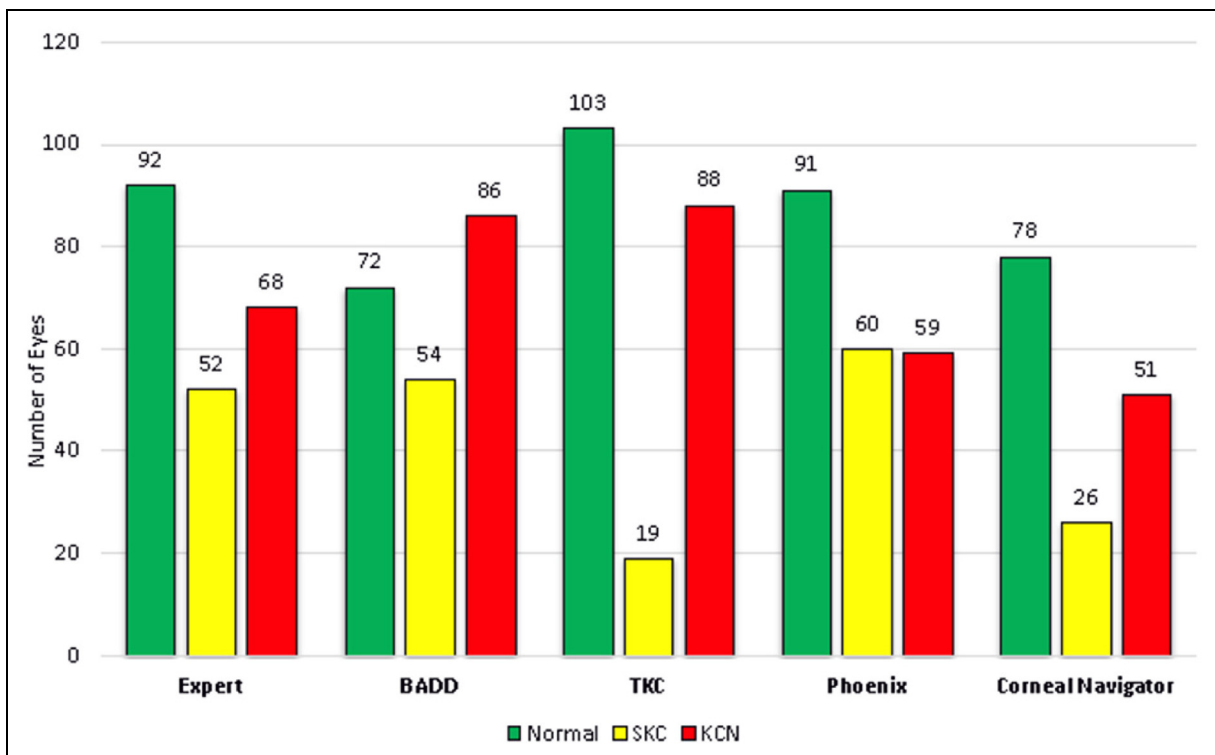


Figure 3. Distribution of normal, subclinical keratoconus (SKCN) and keratoconus (KCN) cases using the four studied classifiers including BADD (Belin/Ambrosio enhanced ectasia total deviation value) and TKC (Topographical Keratoconus Classification) of Pentacam, Phoenix software of Sirius, and Corneal Navigator Classifier of OPD-Scan III, and expert opinion.

detect KCN and normal eyes with higher diagnostic accuracy compared to TKC which correctly identified all normal and KCN eyes, but had a number of false-positive ones. Similarly, Salomao et al.²⁰ compared 226 normal and 44 keratoconic eyes using TKC and reported false-positive results in 7% of normal eyes and false-negative results in 5% of KCN eyes (sensitivity 95.5%, specificity 92.9%). Studying the accuracy of different KCN parameters in the TKC classification system, Wahba et al.²¹ found that pachymetric indices were highly accurate in early KCN cases (TKC grade 1), but their study did not include the assessment of SKCN. In another study by Hidalgo et al.,²² TKC classified all normal eyes correctly, but also placed 13/54 KCN cases and all SKCN cases (23/23) in the N group, while BADD classified fewer SKCN cases as N (17/23 SKCN cases). Similarly, we found that TKC classified a large number of normal eyes correctly (93% of cases), but labeled 30% of SKCN cases as N; BADD classified only 7% of SKCN cases as N. Both algorithms had low sensitivity to the detection of subtle signs of the early form of KCN, however, BADD with better performance than TKC, had a moderate agreement with expert opinion in SKCN detection. Because contrary to TKC that only evaluates the anterior surface of the cornea, BADD provides a more accurate assessment of the cornea based on elevation changes of both corneal surfaces and thickness changes. Previous studies found the high accuracy of BADD in detecting ectatic cases,^{12,23,24} but some BADD rates associated with fewer false negatives may alter these results. The $D > 1.88$ SD (FP < 2.5%, FN < 1%) and $D > 2.69$ SD (FP = 0%) have been suggested for KCN screening and treatment, respectively.²⁵ This rate may vary in different populations and clinical judgments of different experts. In this study, the cut-off value suggested in the Pentacam manual was used.

Among different classifiers, Sirius Phoenix had the highest agreement with expert opinion and high accuracy in diagnosing keratoconic eyes. Also, Phoenix was able to differentiate SKCN eyes from normal and KCN with a higher diagnostic power compared to BADD. In the Phoenix classification method, in addition to elevation and thickness variables, curvature and aberration parameters are used; therefore, it has higher accuracy compared to other classification systems. Few studies have investigated Sirius parameters in the anterior and posterior cornea for detecting ectatic cases from normal eyes. The application of aberration and thickness parameters^{26,27} and certain diagnostic algorithms developed based on support vector machine⁹ as diagnostic parameters has been suggested; however, to the best of our knowledge, there has been no study that directly evaluates the accuracy and diagnostic ability of the Sirius Phoenix classifier. Shetty et al.²⁸ compared the accuracy of Sirius and Pentacam parameters in KCN detection and reported BADD to have the highest accuracy in SKCN detection followed by index of height asymmetry (IHA) and index

of surface variance (ISV) as measured with Pentacam and the back symmetric index (SI_b) with Sirius, however, they did not investigate the accuracy of automatic classification on these two devices. A recent study found that Sirius classifications and BADD were consistent in differentiating SKCN from normal eyes in 76% of the refractive surgery candidates,²⁹ however, their methodology was different from the current study. They first classified the cases with Sirius Phoenix, and then the results were compared with BADD; expert clinical assessment was not used for classification.

In this study, the OPD-Scan III, which evaluates the anterior corneal surface using a neural network, had the lowest diagnostic accuracy of KCN detection compared to other devices. The Corneal Navigator classifier diagnosed 30% of the SKCN eyes correctly and placed a large number of eyes in the OTH group, which represents a poor agreement with expert opinion. A limited number of studies have investigated the OPD-Scan III Corneal Navigator classification system. In one study, Saad et al.³⁰ evaluated 143 patients in three groups of KCN ($n = 31$), forme fruste KCN (FFKCN) ($n = 40$), and N ($n = 72$) using the Orbscan IIz and OPD-Scan III; they suggested that early KCN cannot be detected with Placido-based neural networks, and normal corneas can be recognized using a Corneal Navigator classifier through a topographic pattern. However, direct analysis of Placido mires through the absolute value of the linear regression analysis may increase the diagnostic ability of Placido-based systems in SKCN cases.³¹

AI-based classifiers have shown better performance in differentiating normal eyes from KCN eyes, although they have weaker agreement with expert opinion in detecting SKCN, because they are based on predetermined findings that may not be similar in all devices. Moreover, these algorithms are developed without a definite definition of SKCN cases and small sample sizes.

Pre-refractive surgery screening to prevent iatrogenic ectasia requires accurate identification of SKCN patients, because none of the corneal imaging devices has a definite diagnostic index for KCN diagnosis, especially in early stages. In this case, only the judgment of an expert clinician can be the gold standard, and correct interpretation of different corneal maps can help make sound surgical decisions, because topographic patterns are more helpful in diagnosis compared to the quantitative values of the parameters.

In this study, we consulted three expert corneal specialist to arrive at a valid clinical opinion. However, even an expert physician's diagnosis may not be free of error, and it is necessary to conduct short and long term follow-up studies to further validate results.

Of the limitations of the current cohort, one is the limited number of cases in the groups. In addition, there was no follow up to examine the stability in the seemingly normal cases, disease progression in keratoconic eyes, and ectasia development in cases who received refractive surgery.

In conclusion, among different classification techniques, Sirius Phoenix had the highest agreement with

expert opinion in detecting SKCN and can be helpful in detecting early keratoconus. However, none of these classifiers can replace the clinical experience of expert surgeons and can only be used for screening purposes.

The development of deep learning in AI in the near future may provide a more accurate and timely diagnosis of early forms of keratoconic eyes compatible with clinical expert opinion, and AI-based classifiers may serve as a reliable tool, especially for less experienced practitioners.

Acknowledgements

The authors thank Kazem Amanzadeh MD. and Alireza Moghaddasi MD. for assisting with classifying cases.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship and/or publication of this article.

ORCID iDs

Mehrdad Mohammadpour  <https://orcid.org/0000-0002-9383-6362>

Zahra Heidari  <https://orcid.org/0000-0003-1838-9408>

Hassan Hashemi  <https://orcid.org/0000-0002-2109-0856>

Akbar Fotouhi  <https://orcid.org/0000-0002-6438-6833>

References

- Mohammadpour M, Heidari Z and Hashemi H. Updates on Managements for Keratoconus. *J Curr Ophthalmol* 2018; 30: 110–124.
- Mohammadpour M, Heidari Z, Mirghorbani M, et al. Smartphones, tele-ophthalmology, and VISION 2020. *Int J Ophthalmol* 2017; 10: 1909–1918.
- Stevenson CH, Hong SC and Ogbuehi KC. Development of an artificial intelligence system to classify pathology and clinical features on retinal fundus images. *Clin Exp Ophthalmol* 2019; 47: 484–489.
- Deo RC. Machine learning in medicine. *Circulation* 2015; 132: 1920–1930.
- Smadja D, Touboul D, Cohen A, et al. Detection of subclinical keratoconus using an automated decision tree classification. *Am J Ophthalmol* 2013; 156: 237–246.
- Klyce SD, Karon MD and Smolek MK. Screening patients with the corneal navigator. *J Refract Surg* 2005; 21(5 Suppl): 617–622.
- Kovacs I, Mihaltz K, Kranitz K, et al. Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. *J Cataract Refract Surg* 2016; 42: 275–283.
- Saad A and Gatinel D. Validation of a new scoring system for the detection of early forme of keratoconus. *Int J Kerat Ect Cor Dis* 2012; 2: 100–108.
- Arbelaez MC, Versaci F, Vestri G, et al. Use of a support vector machine for keratoconus and subclinical keratoconus detection by topographic and tomographic data. *Ophthalmology* 2012; 119: 2231–2238.
- Pinero DP, Alio JL, Aleson A, et al. Corneal volume, pachymetry, and correlation of anterior and posterior corneal shape in subclinical and different stages of clinical keratoconus. *J Cataract Refract Surg* 2010; 36: 814–825.
- Belin MW and Ambrósio RJr. Corneal ectasia risk score: statistical validity and clinical relevance. *J Refract Surg* 2010; 26: 238–240.
- Belin MW and Khachikian SS. Keratoconus/ectasia detection with the oculus pentacam: Belin/Ambrósio enhanced ectasia display. *Highlights Ophthalmol* 2007; 35: 5–12.
- Muckenhirn D. [Fitting of aspheric contact lenses in keratoconus under consideration of the geometric-optical conditions of the cornea]. *Neues Optik J* 1984; 5: 87–94.
- Belin MW, Villavicencio OF and Ambrosio RJr. Tomographic parameters for the detection of keratoconus: suggestions for screening and treatment parameters. *Eye Contact Lens* 2014; 40: 326–330.
- Mohammadpour M. and Heidari Z. Pentacam. In: Mohammadpour M (eds). *Diagnostics in Ocular Imaging*. Switzerland: Springer Cham, 2021, pp.65–162.
- Rabinowitz YS and McDonnell PJ. Computer-assisted corneal topography in keratoconus. *Refract Corneal Surg* 1989; 5: 400–408.
- Toprak I, Vega A, Alió Del Barrio JL, et al. Diagnostic value of corneal epithelial and stromal thickness distribution profiles in forme fruste keratoconus and subclinical keratoconus. *Cornea* 2021; 40: 61–72.
- Maeda N, Klyce SD, Smolek MK, et al. Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci* 1994; 35: 2749–2757.
- Viera AJ and Garrett JM. Understanding interobserver agreement: the kappa statistic. *Fam Med* 2005; 37: 360–363.
- Salomao MQ, Guerra FP, Ramos IC, et al. Accuracy of topometric indices for distinguishing between keratoconic and normal corneas. *Int J Kerat Ect Cor Dis* 2013; 2: 108–112.
- Wahba SS, Roshdy MM, Elkitkat RS, et al. Rotating Scheimpflug Imaging Indices in Different Grades of Keratoconus. *J Ophthalmol* 2016; 2016: 6392472.
- Ruiz Hidalgo I, Rozema JJ, Saad A, et al. Validation of an Objective Keratoconus Detection System Implemented in a Scheimpflug Tomographer and Comparison With Other Methods. *Cornea* 2017; 36: 689–695.
- Correia FF, Ramos I, Lopes B, et al. Topometric and tomographic indices for the diagnosis of keratoconus. *Int J Kerat Ect Cor Dis* 2012; 1: 92–99.
- Shajari M, Jaffary I, Herrmann K, et al. Early Tomographic Changes in the Eyes of Patients With Keratoconus. *J Refract Surg* 2018; 34: 254–259.
- Villavicencio OF, Gilani F, Henriquez MA, et al. Independent population validation of the Belin/Ambrosio enhanced ectasia display: implications for keratoconus studies and screening. *Int J Kerat Ect Cor Dis* 2014; 3: 1–8.
- Safarzadeh M and Nasiri N. Anterior segment characteristics in normal and keratoconus eyes evaluated with a combined Scheimpflug/Placido corneal imaging device. *J Curr Ophthalmol* 2016; 28: 106–111.

27. Heidari Z, Mohammadpour M, Hashemi H, et al. Early diagnosis of subclinical keratoconus by wavefront parameters using Scheimpflug, Placido and Hartmann-Shack based devices. *Int Ophthalmol* 2020; 40: 1659–1671.
28. Shetty R, Rao H, Khamar P, et al. Keratoconus screening indices and their diagnostic ability to distinguish normal From ectatic corneas. *Am J Ophthalmol* 2017; 181: 140–148.
29. Zhang Y, Chen YG, Yang HY, et al. [Analysis on the role of Sirius combined topography and tomography system in screening for suspect keratoconus]. *Zhonghua Yan Ke Za Zhi* 2018; 54: 33–38.
30. Saad A and Gatinel D. Topographic and tomographic properties of forme fruste keratoconus corneas. *Invest Ophthalmol Vis Sci* 2010; 51: 5546–5555.
31. Ramos-Lopez D, Martinez-Finkelshtein A, Castro-Luna GM, et al. Screening subclinical keratoconus with placido-based corneal indices. *Optom Vis Sci* 2013; 90: 335–343.